

3.1.6 Activity: Students Attendance Program

حاضری کا پروگرام طلبہ کے نام ایک فہرست میں رکھتا ہے، ہر نام کے لیے پوچھتا ہے، اور yes پر شمار کار (Counter) ایک بڑھاتا ہے۔

CS XII

Chapter 3: Programming Fundamentals

Lecture 07 of 17

3.1.6 Activity: Students Attendance Program

LEARN IT

Teachers need a quick way to mark who is in class. This program stores names in a **list**, asks about each student, counts how many said yes, and prints a summary with `len()`.

Store names in a list → Ask present? for each → Count yes → Print totals

```
students = ["Ali", "Sara", "Ayesha"]
present_count = 0
print("--- Attendance System ---\n")
for student in students:
    status = input(f"Is {student} present? (yes/no): ").lower()
    if status == "yes":
        present_count += 1
print("\n--- Attendance Summary ---")
print("Total Students:", len(students))
print("Students Present:", present_count)
print("Students Absent:", len(students) - present_count)
```

WORKED EXAMPLE

Sample run. Ali yes, Sara no, Ayesha yes. `.lower()` means YES or Yes still counts as present.

OUTPUT

```
--- Attendance System ---  
  
Is Ali present? (yes/no): yes  
Is Sara present? (yes/no): no  
Is Ayesha present? (yes/no): yes  
  
--- Attendance Summary ---  
Total Students: 3  
Students Present: 2  
Students Absent: 1
```

Student	Typed answer	After <code>.lower()</code>	<code>present_count</code>
Ali	yes	yes → add 1	1
Sara	no	not yes	1
Ayesha	yes	yes → add 1	2

Absent = `len(students) - present_count` = 3 - 2 = 1. The counter must start at 0 **before** the loop. If you put `present_count = 0` inside the loop, the count would reset for every student.

REMEMBER IT

Counter pattern: start at 0, loop, add 1 when a condition is true. You will reuse this in the word-frequency program (add 1 per word) and the grade tracker (add each mark).

COMMON MISTAKE

Only the exact string "yes" after `.lower()` counts as present. "y" or "present" is treated as absent. In a longer program you could also accept "y"; this activity keeps the rule simple.

اردو میں سمجھیں

حاضری کا پروگرام طلبہ کے نام ایک فہرست میں رکھتا ہے، ہر نام کے لیے پوچھتا ہے، اور yes پر شمار کار (Counter) ایک بڑھاتا ہے۔ کل طلبہ `len(students)` سے، غیر حاضر کل - حاضر سے نکلتے ہیں۔ شمار کار لوپ سے پہلے صفر ہونا چاہیے۔

CHECK YOURSELF

1. If all three students type `no`, what is printed for present and absent?
2. Why is `.lower()` used on the answer?

Answers: 1. Present 0, Absent 3. 2. So that YES, Yes and yes all match the string "yes".

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. A sensible structure for present/absent flags by name is:

- a. dict name→status or two lists/sets
- b. A pie without data
- c. A NAND net
- d. An OSI capture

MCQ 2. You should still:

- a. Validate input (yes/no)
- b. Trust any string
- c. Use eval
- d. Open the OS

MCQ 3. Printing a report uses:

- a. A loop over the collection
- b. A single print of the object id
- c. TCP
- d. Figma

MCQ 4. This activity combines:

- a. DS + input + selection + repetition
- b. Only HCI colours
- c. Only ER
- d. Only Big O proofs

MCQ 5. A set of absentees is useful to:

- a. Unique names, fast membership
- b. Keep duplicate absences as a stack
- c. FIFO print jobs
- d. Store key-value marks

MCQ 6. Never store:

- a. Passwords in the same toy script unprotected
- b. Names
- c. P/A flags
- d. Dates

Extra short questions

1. Outline the attendance program's data and loop.
2. Two Python DS options.
3. How do you count presents?
4. One improvement for XII files.

Extra long question

1. Write a complete small attendance program for 3 names using a dict, print a tidy table, and a sample run.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. For each student: ask status; store; then print counts.
2. dict[str,str] or present_set/absent_set.
3. Counter in the loop or list.count / sum(1 for ...).
4. Save to a text file with with-open.

Long

1. Working code, output, and one validation (only P/A). Mention a logic error if you forget to store.

Next lecture: 3.1.7 Activity: Word Frequency Counter